

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: -6 ; $+4$
- Ex 1b: 9 , -15 , 0
- Ex 2a: -3 , $+3$; back where it started
- Ex 2b: -8 (8 units from 0)
- Ex 3a: -25
- Ex 3b: 7 and -7

Easy (drill)

- | | | |
|-----------|-----------|-----------|
| 1. -55 | 11. 39 | 21. 61 |
| 2. -27 | 12. 10 | 22. -12 |
| 3. -45 | 13. 51 | 23. 43 |
| 4. 6 | 14. -31 | 24. 65 |
| 5. 21 | 15. -16 | 25. 28 |
| 6. 19 | 16. 7 | 26. 67 |
| 7. -3 | 17. 87 | 27. -73 |
| 8. -48 | 18. 30 | 28. 36 |
| 9. -21 | 19. -50 | |
| 10. -42 | 20. 69 | |

Medium (drill)

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|----------|-----------|------------------|
| 1. -22 | 9. -7 | 17. -54 , 54 |
| 2. 15 | 10. -30 | 18. 83 , -83 |
| 3. -8 | 11. -16 | 19. -46 , 46 |
| 4. -25 | 12. -9 | 20. 63 , -63 |
| 5. 19 | 13. 13 | 21. -6 , 6 |
| 6. 21 | 14. -19 | 22. 20 , -20 |
| 7. 21 | 15. 24 | |
| 8. 28 | 16. 16 | |

Challenging

- No, $-0 = 0$
- -18
- -2 , -1 , 0 , 1 , 2

Extension

- 11 ; -11
- No: if n is negative, $-n$ is positive (e.g. $n = -3$, $-n = 3$)

Exit ticket

- Level 1: 15
- Level 2: -9 ; 9
- Level 3: 31 , because the opposite of 31 is -31

1.02 The number line (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: dots at -5 , -2 , 0 , 3
- Ex 1b: -2 , -1 , 0 , 1
- Ex 2a: dots at -4 , -1 , 2 , 5
- Ex 2b: -5
- Ex 3a: scale in 5 s; dots at -30 , -15 , 10
- Ex 3b: 9

Easy (drill)

- | | | |
|---------|----------|----------|
| 1. 14 | 10. 12 | 19. 11 |
| 2. 12 | 11. 15 | 20. 13 |
| 3. 4 | 12. 12 | 21. 13 |
| 4. -3 | 13. 7 | 22. 3 |
| 5. 3 | 14. 10 | 23. 6 |
| 6. 12 | 15. 10 | 24. 12 |
| 7. 3 | 16. -2 | 25. -6 |
| 8. 2 | 17. -1 | 26. -5 |
| 9. 12 | 18. 5 | 27. -6 |

28. 5

Medium (drill)

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|---------|----------|----------|
| 1. 22 | 9. 7 | 17. 9 |
| 2. 2 | 10. 8 | 18. -3 |
| 3. 36 | 11. 26 | 19. 3 |
| 4. 17 | 12. 18 | 20. -4 |
| 5. 20 | 13. 31 | 21. 8 |
| 6. 4 | 14. 1 | 22. -5 |
| 7. 3 | 15. 14 | 23. -8 |
| 8. 15 | 16. 13 | 24. 0 |

Challenging

- scale -40 to 40 in 10 s with dots
- -3 , -2 , -1 , 0 , 1 , 2

Extension

- 54
- -6

Exit ticket

- Level 1: -2 , -1 , 0
- Level 2: -8
- Level 3: -3

1.03 Comparing and ordering integers

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: False; false
- Ex 1b: -6 , -1 , 0 , 3 , 5
- Ex 2a: $>$
- Ex 2b: 8 , 3 , 0 , -2 , -15
- Ex 3a: -5 , $-(2)$, 1 , $-(-3)$
- Ex 3b: -11 , -10 , -9

Easy (drill)

- | | | |
|--------|---------|---------------------------------|
| 1. $>$ | 9. $<$ | 17. -10 , -5 , 6 , 8 |
| 2. $<$ | 10. $<$ | 18. -15 , -12 , -3 , 10 |
| 3. $>$ | 11. $<$ | 19. -16 , 2 , 15 , 19 |
| 4. $<$ | 12. $<$ | 20. -21 , -13 , -3 , 14 |
| 5. $>$ | 13. $>$ | 21. -25 , -12 , -7 , -2 |
| 6. $<$ | 14. $<$ | 22. -13 , -8 , 5 , 12 |
| 7. $<$ | 15. $>$ | 23. -6 , 0 , 2 , 3 |
| 8. $>$ | 16. $<$ | 24. -13 , -9 , 3 , 12 |

Medium (drill)

- | | | |
|----------|-----------|---|
| 1. false | 8. false | 15. true |
| 2. true | 9. true | 16. false |
| 3. false | 10. true | 17. 22 , 21 , 15 , 3 , -10 |
| 4. true | 11. false | 18. 27 , 12 , -24 , -25 , -26 |
| 5. false | 12. false | 19. 28 , 20 , 19 , 13 , 2 |
| 6. true | 13. true | 20. 28 , 24 , -4 , -24 , -39 |
| 7. true | 14. true | |

Challenging

- e.g. -6 , -7 , -8
- -8
- -1000 , -100 , -10 , -1 , 0

Extension

- -4 , $-(-2)$, 3 , $-(-4)$, $-(-9)$
- e.g. 0 and 2 (any integer from -3 to 2)

Exit ticket

- Level 1: False

2. Level 2: 2, 0, -1, -5
3. Level 3: -20, -12, 0, 5, -(-8)

28. 11 30. 2 32. 2
29. 14 31. 8

1.04 Adding integers (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: -1
2. Ex 1b: -10; 4
3. Ex 2a: -11
4. Ex 2b: 1
5. Ex 3a: 5°C
6. Ex 3b: -13

Easy (drill)

- | | | |
|--------|---------|---------|
| 1. 5 | 12. 3 | 23. -12 |
| 2. 6 | 13. 3 | 24. 0 |
| 3. -2 | 14. 7 | 25. -6 |
| 4. 5 | 15. -5 | 26. 7 |
| 5. 2 | 16. 9 | 27. -17 |
| 6. -4 | 17. -18 | 28. -18 |
| 7. 6 | 18. -8 | 29. -2 |
| 8. 5 | 19. 2 | 30. -13 |
| 9. 0 | 20. 8 | 31. -2 |
| 10. -4 | 21. 4 | 32. 1 |
| 11. -1 | 22. 5 | |

Medium (drill)

- | | | |
|--------|----------|---------|
| 1. 7 | 8. 17 | 15. 125 |
| 2. -15 | 9. 8 | 16. 18 |
| 3. 7 | 10. 4 | 17. 85 |
| 4. -10 | 11. 0 | 18. 6 |
| 5. -18 | 12. -2 | 19. 25 |
| 6. -10 | 13. -145 | 20. -31 |
| 7. -6 | 14. 132 | |

Challenging

1. 5°C
2. level 5
3. -\$15

Extension

1. -12
2. 7

Exit ticket

1. Level 1: 3
2. Level 2: -7
3. Level 3: 1°C

1.05 Subtracting integers (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: -4; -12
2. Ex 1b: 7; 4
3. Ex 2a: -12
4. Ex 2b: 6
5. Ex 3a: 15°C
6. Ex 3b: -8

Easy (drill)

- | | | |
|--------|---------|--------|
| 1. -14 | 10. -3 | 19. 11 |
| 2. -5 | 11. -3 | 20. 16 |
| 3. -11 | 12. -10 | 21. 2 |
| 4. -11 | 13. 2 | 22. -4 |
| 5. -10 | 14. -17 | 23. 1 |
| 6. -18 | 15. -8 | 24. 0 |
| 7. -19 | 16. -15 | 25. 2 |
| 8. -14 | 17. -6 | 26. 7 |
| 9. -4 | 18. 5 | 27. 9 |

Medium (drill)

- | | | |
|--------|---------|---------|
| 1. -25 | 9. 32 | 17. 7 |
| 2. -15 | 10. 30 | 18. 22 |
| 3. 19 | 11. 53 | 19. 12 |
| 4. -1 | 12. 43 | 20. 8 |
| 5. -13 | 13. -34 | 21. -13 |
| 6. -48 | 14. 21 | 22. 4 |
| 7. 14 | 15. 40 | |
| 8. 56 | 16. 45 | |

Challenging

1. 16°C
2. 25 m
3. -75 m (75 m deep)

Extension

1. -8
2. -10

Exit ticket

1. Level 1: -9
2. Level 2: 13
3. Level 3: 19°C

1.06 Multiplying integers (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: -18; -10
2. Ex 1b: 48; -24
3. Ex 2a: -42
4. Ex 2b: 40
5. Ex 3a: -8
6. Ex 3b: Positive: 100 negatives make 50 pairs

Easy (drill)

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|---------|---------|-------|
| 1. -18 | 12. -5 | 23. - |
| 2. 9 | 13. 3 | 24. + |
| 3. -40 | 14. -30 | 25. - |
| 4. -42 | 15. -6 | 26. - |
| 5. 30 | 16. -7 | 27. + |
| 6. 80 | 17. + | 28. + |
| 7. -14 | 18. + | 29. - |
| 8. -18 | 19. + | 30. + |
| 9. 48 | 20. + | 31. - |
| 10. -24 | 21. - | 32. - |
| 11. 72 | 22. - | |

Medium (drill)

- | | | |
|---------|---------|--------|
| 1. -100 | 8. -20 | 15. 2 |
| 2. -20 | 9. 6 | 16. -3 |
| 3. -6 | 10. -8 | 17. 3 |
| 4. 4 | 11. -30 | 18. -3 |
| 5. -24 | 12. -4 | 19. 6 |
| 6. -150 | 13. -8 | 20. -3 |
| 7. -36 | 14. -6 | |

Challenging

1. -28 m (28 m deep)
2. -7
3. Negative: 2 pairs make positive, 1 negative left

Extension

1. -32
2. 8 and -3

Exit ticket

1. Level 1: -42

2. Level 2: 72
3. Level 3: -60

1.07 Dividing integers (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 3; -3
2. Ex 1b: 2
3. Ex 2a: -3
4. Ex 2b: 8
5. Ex 3a: -\$12
6. Ex 3b: 54

Easy (drill)

1. 6	12. 3	23. -6
2. -6	13. -3	24. 6
3. -1	14. -10	25. 2
4. 4	15. -7	26. 3
5. -7	16. 6	27. 10
6. 8	17. 8	28. 7
7. -9	18. -12	29. -1
8. -4	19. -9	30. 1
9. -5	20. -2	31. 9
10. -7	21. 11	32. 2
11. 6	22. -5	

Medium (drill)

1. 3	8. -1	15. -3
2. -3	9. -5	16. -8
3. 4	10. -4	17. 2
4. -1	11. 2	18. -1
5. -4	12. 4	19. 9
6. -5	13. -2	20. 4
7. 1	14. -5	

Challenging

1. -\$21
2. -3°C each hour
3. -35

Extension

1. -60
2. -5

Exit ticket

1. Level 1: -4
2. Level 2: 7
3. Level 3: -4

1.08 Order of operations with integers

(booklet p. 31)

WE DO examples (teacher)

1. Ex 1a: -12
2. Ex 1b: -4
3. Ex 2a: -18
4. Ex 2b: 7
5. Ex 3a: $1 \div (-3) = -1/3$
6. Ex 3b: 0

Easy (drill)

1. 7	9. -29	17. 0
2. 35	10. -41	18. 32
3. -12	11. 19	19. -14
4. 8	12. -17	20. 4
5. 6	13. -10	21. -18
6. 6	14. 4	22. -25
7. 1	15. 90	23. 9
8. 3	16. -5	24. -5

Medium (drill)

1. -5	6. 6	11. 15
2. 26	7. 0	12. 35
3. 0	8. 15	13. -28
4. -1	9. 21	14. 22
5. -9	10. 16	15. 18

Challenging

1. -4
2. $-1/3$
3. 4

Extension

1. $(8 - 12) \div (-4 + 2) = 2$
2. e.g. $-3 \times 4 - 5 = -17$

Exit ticket

1. Level 1: 1
2. Level 2: 15
3. Level 3: 12

1.09 Integers on a calculator (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: -9; -6
2. Ex 1b: 5
3. Ex 2a: -12
4. Ex 2b: 87
5. Ex 3a: -52
6. Ex 3b: 4

Easy (drill)

1. -33	12. 357	23. -400
2. 397	13. -374	24. 1 230
3. 469	14. 576	25. 136
4. -167	15. 304	26. -246
5. 667	16. 19	27. -1 530
6. 19	17. 456	28. 740
7. 55	18. -170	29. 322
8. -80	19. -33	30. -210
9. 213	20. -168	31. 240
10. 97	21. -238	32. 924
11. -172	22. 154	

Medium (drill)

1. 54	9. 24	17. 85
2. -1	10. -16	18. 405
3. 58	11. -17	19. 2 232
4. 12	12. 4	20. 297
5. -58	13. -5	21. 1 003
6. -27	14. -16	22. -1 014
7. 42	15. 21	
8. 2	16. -5	

Challenging

1. ≈ -1000 ; -1029
2. ≈ -20 ; -20.5
3. The calculator squares 6 first, then makes it negative.
Enter $(-6)^2$

Extension

1. 5
2. -5

Exit ticket

1. Level 1: -45
2. Level 2: 5
3. Level 3: 5

1.10 Integer problems (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: 16°C
2. Ex 1b: $-\$45$
3. Ex 2a: level -3
4. Ex 2b: $\$85$
5. Ex 3a: $\$820$
6. Ex 3b: -12 m

Easy (drill)

- | | | |
|-------------------------|-------------------------|-------------|
| 1. 1°C | 7. 12°C | 13. $-\$31$ |
| 2. 0°C | 8. 17°C | 14. $\$183$ |
| 3. 26°C | 9. $\$61$ | 15. $\$54$ |
| 4. -6°C | 10. $-\$57$ | 16. $\$25$ |
| 5. 4°C | 11. $\$106$ | |
| 6. 16°C | 12. $\$111$ | |

Medium (drill)

1. 191 m
2. 138 m

3. 393 m
4. 341 m
5. 114 m
6. 104 m
7. 225 m
8. 247 m
9. $-80 + 3 \times 4 = -68\text{ m}$
10. $-95 + 2 \times 2 = -91\text{ m}$

Challenging

1. $26 \times 80 - 6 \times 210 = \820
2. $35 - 27 = 8$

Extension

1. 70 minutes
2. e.g. It was 4°C and fell 12°C

Exit ticket

1. Level 1: 4°C
2. Level 2: 18 m
3. Level 3: -7